PROJECT CHARTER

Official project name: DentaLead

Project sponsor: Ministry of Health

Project manager: Arwa Mohammed Hussien

Purpose of the project: To help dental students in universities easily find patients to practice on, and to help patients reach doctors more efficiently, so we can solve the problem of students' difficulty in accessing patients and getting proper clinical training, as well as patients' difficulty in receiving timely dental care.

Business case

DentaLead is an application designed to make clinical training easier for dental students and to provide better care for patients.

The problem is that Dental students at university clinics often struggle to find patients for their clinical training. On the other hand, patients face long waiting times and difficulty booking appointments or accessing care efficiently.

The app allows dental students to create accounts, and whenever a patient visits the university clinic and registers, their case details will appear in the students' case list. Students can then book the case and schedule an appointment.

Patients can also easily register their information through computers at the reception area in the university hospital, allowing them to book appointments more efficiently and attend at the scheduled time — instead of waiting for long periods or coming back multiple times.

Key deliverables

- DentLead Mobile Application
- Patient Registration System
- Appointment Booking & Scheduling Feature

Project milestones

- **Requirements review** : Requirements specifications are complete, correct, approved and suitable. By 20/11/2026
- **Critical design review**: Detailed designs fully implement the system architecture, are approved and are suitable for input into the development of code. By 1/1/2026
- **System test review:** The software product has passed system testing and is suitable for input into acceptance testing.by 23/1/2026
- **Deployment & Documentation Review:** The application has been successfully deployed to Android and iOS platforms, and all required user manuals, technical documentation. By 7/2/2026
- **Project Closure Review**: Fina review a, submitting final report and handover the software to the sponser.by 18/2/2026

Project resources

- Staff :
  2 backend developers (for android and IOS), 2 frontend developers (for android and IOS), UI/UX Designer, Designer, DB Dev, DB Designer, System Analyst, Researcher, tester, QA Engineer, Tech Writer

- Hardware : 12 laptops , 1 server

- Software and Software licenses :

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1. **Frontend (Android & iOS)**

a) Android Studio – for Android app development (free, open-source)
b) Xcode – for iOS app development (free, macOS only)
c) React Native / Flutter / Swift / Kotlin IDEs – depends on your tech stack
d) License: Most IDEs are free; Xcode requires a macOS device. Publishing on App Store requires Apple Developer Program.

2. **Backend**

a) Node.js / Python / Java / .NET depending on backend tech
b) IDE/Editor: VS Code (free) or IntelliJ IDEA (commercial for advanced features)
c) Database tools: pgAdmin, MySQL Workbench, DBeaver (free/open-source)
d) API Testing Tools: Postman (free / paid version for team collaboration)
e) Deployment and Management Software

3. **Database**

a) **Database server**: PostgreSQL, MySQL, SQL Server, or MongoDB (depending on design)
b) **DB Design Tools**:
c) dbdiagram.io (free/paid)
d) MySQL Workbench (free)
e) Lucidchart / Draw.io for ER diagrams (Draw.io free, Lucidchart paid for advanced features)

4. **UI/UX Design**

a) **Figma** – free tier available; paid for team collaboration and versioning
b) **Adobe XD / Sketch** – optional; paid licenses
c) **Illustrator / Photoshop** – optional for graphics (paid)

5. **QA & Testing**

a) **Manual Testing Tools**: TestRail (paid), Zephyr (paid), Jira Test Management
b) **Automation Testing**: Selenium (free), Appium (free)
c) **Bug tracking / Project management**: Jira (paid), Trello (free/paid), Asana

6. **Documentation & Tech Writing**

a) Microsoft Office (Word, Excel, PowerPoint) – paid license or Office 365 subscription
b) Google Workspace – alternative (paid/free options)
c) Confluence – for collaborative technical documentation (paid)

7. **Version Control / Collaboration**

a) Git (free)
b) GitHub / GitLab / Bitbucket – free tiers available, paid for private repos and team features
c) Slack / Microsoft Teams / Zoom – communication (some free tiers)

8. **Deployment / CI-CD**

a) CI/CD Tools: GitHub Actions (free tier), Jenkins (free), CircleCI (free/paid)
b) Cloud Hosting: AWS, Azure, GCP, Firebase – free tier / paid depending on usage
c) App Store / Play Store Publishing:
d) Apple Developer Program
e) Google Play Console

Budget

- budget allocated for this project is 100000 L.E

Constraints

- this project must be delivered within the specified budget .
- This project must be delivered on 18/2/2026
- It must have an easy and catchy UI/UX

High level risks

- Low User Adoption
- Technical Issues
- Privacy and Security Risks
- Scope creep: uncontrolled changes and continuous growth of the scope or scope is ill defined
- Project team misunderstands requirements
- Dependency on University Infrastructure

DENTALEAD PROJECT SCOPE

October 20, 2025

OVERVIEW

1. Product Scope Description

DentaLead is a mobile and web application designed to make clinical training for dental students easier and to improve patients' access to care.

The system provides an efficient connection between students and patients through:

- A platform that allows patients to register their information upon visiting the university dental clinic.
- Displaying patient cases in dental students' case lists in the platform.
- Allowing students to book cases and schedule appointments.
- A well-structured appointment scheduling system to minimize waiting times.
- A simple and user-friendly UI/UX interface for both patients and students.
- Compatibility with **mobile devices** (Android and iOS).

The main purpose of this project is to solve the problem of students struggling to access patients for clinical training and patients facing long waiting times, by creating an efficient, accessible, and organized system.

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2. Product Acceptance Criteria

The project will be considered successfully completed and accepted by the **Ministry of Health** if:

- The application functions properly on Android, and iOS devices.
- Patients can register their information easily through the hospital reception system.
- Students can view and book patient cases without errors.
- The appointment system runs smoothly and reliably.
- All project milestones (requirements, design, testing, and deployment) are completed and approved on time.
- The Ministry of Health provides final approval after user testing.
- The system meets security and privacy requirements.

3. Project Deliverables

- *Fully developed **DentaLead** mobile application.*
- *Patient Registration System.*
- *Appointment Booking and Scheduling features.*
- *User documentation (User Guide and FAQs).*
- *Project documentation (requirements, design, and testing reports).*
- *Deployment package and maintenance plan.*

4. Project Exclusions

The following are **not included** in the project scope:

- Integration with external non-university hospital databases.
- Live consultation or telemedicine features.
- Commercial marketing outside target universities.

5. Project Constraints

- Total budget is limited to **100,000 L.E.**
- The project must be delivered by **18/2/2026**.
- The application must have an easy and attractive UI/UX.
- Compliance with **Ministry of Health** security and privacy standards is mandatory.

6. Project Assumptions

- All required software tools and licenses will be available during development.
- The Ministry of Health will provide and approve all educational and regulatory content.
- Developers will have access to the required hardware (12 laptops and 1 server).
- The app will be published under the Ministry of Health’s official account.
- Internet connectivity will be stable during development and deployment, with the option to use offline mode if needed.

DENTALEAD SYSTEM -- WORK BREAKDOWN STRUCTURE (WBS)

Level 1: 1.0 DentaLead System

Level 2: Major Phases

1.1 System Analysis & Planning
1.2 System Design
1.3 System Development
1.4 System Testing & Validation
1.5 Deployment & Documentation
1.6 Project Closure

Level 3: Tasks under Each Phase

1.1 System Analysis & Planning

- 1.1.1 Gather requirements (patient registration, booking system)
- 1.1.2 Identify hardware and software needs
- 1.1.3 Define database schema
- 1.1.4 Create project schedule and assign tasks

1.2 System Design

- 1.2.1 Design user interface and user experience (UI/UX)
- 1.2.2 Design database tables and relationships
- 1.2.3 Prepare prototypes
- 1.2.4 Review and approval of design

1.3 System Development

- 1.3.1 Frontend development for web and mobile
- 1.3.2 Backend development and API integration
- 1.3.3 Implement patient registration system
- 1.3.4 Implement appointment booking feature

1.4 System Testing & Validation

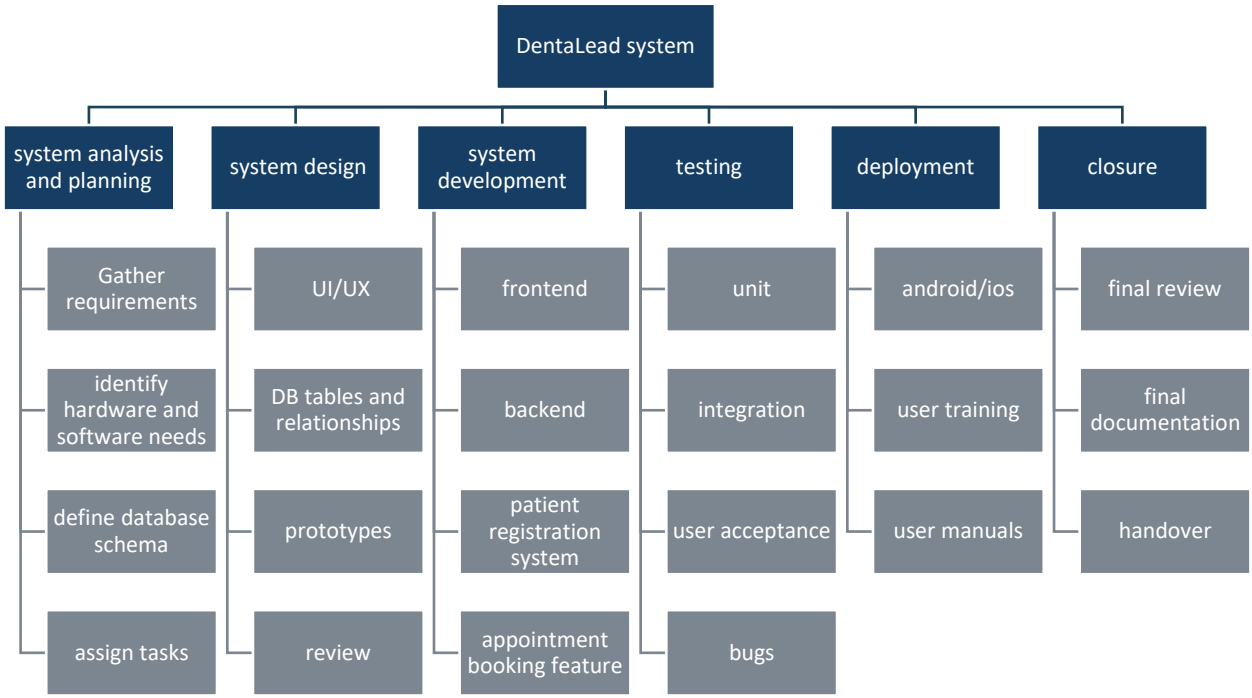
- 1.4.1 Unit testing (individual functions)
- 1.4.2 Integration testing (frontend + backend)
- 1.4.3 User acceptance testing (Ministry of Health approval)
- 1.4.4 Bug fixing and optimization

1.5 Deployment & Documentation

- 1.5.1 Deploy app to web, Android and iOS platforms
- 1.5.2 Conduct user training for students and staff
- 1.5.3 Prepare user manuals and technical documentation

1.6 Project Closure

- 1.6.1 Final review meeting and approval
- 1.6.2 Submit final report and documentation
- 1.6.3 Handover to Ministry of Health



WBS DICTIONARY

Code	Task Description	Assigned People	No. of Resources	Budget (LE)	Milestone (Date)
1.1.1	Gather requirements	Omar Khaled (System Analyst), Sarah Fathy (Researcher)	2	5000	1/11/2025
1.1.2	Identify hardware and software needs	Omar Khaled (System Analyst)	1	2500	6/11/2025
1.1.3	Define database schema	Mariam Said (DB Designer)	1	1500	11/11/2025
1.1.4	Create project schedule and assign tasks	Arwa Mohammed (Project Manager)	1	3000	16/11/2025
1.2.1	Design UI/UX	Hossam Magdy (UI/UX Designer),	1	4000	21/11/2025
1.2.2	Design database tables and relationships	Mariam Said (DB Designer)	1	1500	21/11/2025
1.2.3	Prepare prototypes	Jana Ali (Designer), Youssef Kamal (Frontend Dev)	2	4000	28/11/2025
1.2.4	Review and approve design	Arwa Mohammed (PM), Omar Khaled (System Analyst)	2	3000	8/12/2025
1.3.1	Frontend development for mobile	Youssef Kamal, Reem Tamer (Frontend Developers)	2	8000	11/12/2025

1.3.2	Backend development and API integration	Mariam Said (DB Dev), Ali Hassan (Backend Dev)	2	10000	11/12/2025
1.3.3	Implement patient registration system	Reem Tamer (Frontend Dev), Badr Ali (Backend Dev), Hossam Magdy (UI/UX Designer),	3	10000	21/12/2025
1.3.4	Implement appointment booking feature	Ali Hassan (Backend Dev)	1	2000	27/12/2025
1.4.1	Unit testing (individual functions)	Rania Fathy (Tester), Ahmed Salah (QA Engineer)	2	6000	2/1/2026
1.4.2	Integration testing (frontend + backend)	Rania Fathy (Tester), Ahmed Salah (QA Engineer)	2	6000	7/1/2026
1.4.3	User acceptance testing	Ministry of Health team (3 reviewers)	3	1500	7/1/2026
1.4.4	Bug fixing and optimization	Developers team	2	5000	14/1/2026
1.5.1	Deploy app to web, Android and iOS	Ali Hassan (Dev), Arwa Mohammed(PM)	2	5000	24/1/2026
1.5.2	Conduct user training	Sarah Fathy (Trainer)	1	1000	31/1/2026
1.5.3	Prepare user manuals and technical documentation	Mariam Said (Tech Writer), Jana Ali (Designer)	2	4000	5/2/2026

1.6.1	Final review meeting and approval	Arwa Mohammed (PM), Ministry of Health Rep.	2	4000	8/2/2026
1.6.2	Submit final report and documentation	PM + Team	3	10000	13/2/2026
1.6.3	Handover to Ministry of Health	PM + IT Admin	2	3000	**18/2/2026**

Responsibility Matrix

WBS Code & Task	Arwa M. (PM)	Omar K. (System Analyst)	Sarah F. (Researcher)	Mariam S. (DB Dev)	Hossam M. (UI/UX)	Jana A. (Designer)	Youssef K. (Front end)	Reem T. (Mobile)	Ali H. (Back end)	Rania F. (Tester)	Ahmed S. (QA)	MOH Team
1.1.1 Gather requirements	S	S	R	–	–	–	–	–	–	–	–	–
1.1.2 Identify hardware & software needs	–	R	–	–	–	–	–	–	–	–	–	–
1.1.3 Define database schema (SQLite)	–	S	–	R	–	–	–	–	–	–	–	–
1.1.4 Create project schedule & assign tasks	R	–	–	–	–	–	–	–	–	–	–	–
1.2.1 Design UI/UX	–	–	–	–	R	S	–	–	–	–	–	–

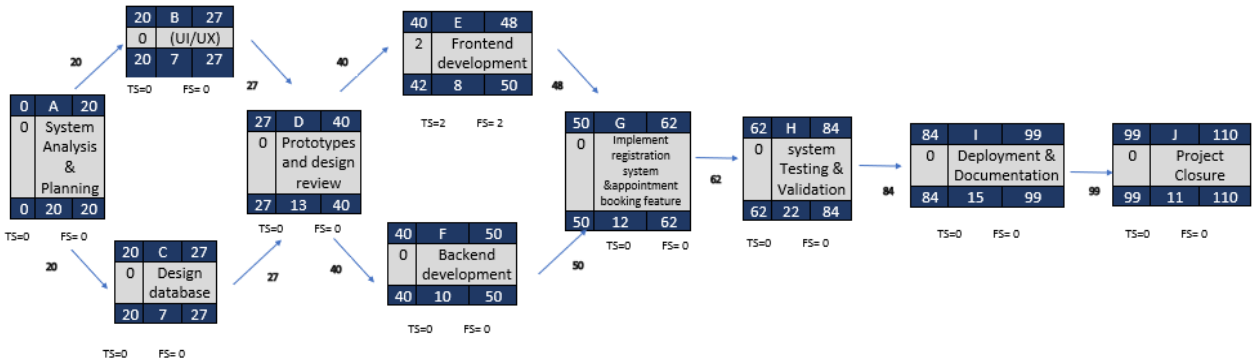
1.2.2 Design database tables & relationships	–	S	–	R	–	–	–	–	–	–	–	–
1.2.3 Prepare prototypes	–	–	–	–	R	S	S	–	–	–	–	–
1.2.4 Review & approval of design	R	S	–	S	–	–	–	–	–	–	–	–
1.3.1 Front end development (Flutter)	–	–	–	–	S	–	S	R	–	–	–	–
1.3.2 Backend development (SQLite)	–	–	–	R	–	–	–	–	S	–	–	–
1.3.3 Implement patient registration system	–	–	–	–	S	–	–	R	S	–	–	–
1.3.4 Implement appointment booking feature	–	–	–	–	–	–	–	–	R	–	–	–
1.4.1 Unit testing (individualfunctions)	–	–	–	–	–	–	S	S	S	S	R	–
1.4.2 Integration testing (Flutter+SQLite)	–	–	–	S	–	–	S	–	S	S	R	–
1.4.3 User acceptance	S	–	–	–	–	–	–	–	–	–	S	R

testing (MOH approval)												
1.4.4 Bug fixing & optimization	–	–	–	S	–	–	R	S	S	–	S	–
1.5.1 Deploy app (Android/iOS)	S	–	–	S	–	–	S	S	R	–	–	–
1.5.2 Conduct user training	S	–	R	–	–	–	–	–	–	–	–	S
1.5.3 Prepare manuals & documentation	S	–	–	R	–	S	–	–	–	–	–	–
1.6.1 Final review & approval meeting	R	S	–	–	–	–	–	–	–	–	–	S
1.6.2 Submit final report & documentation	R	S	–	S	–	–	–	–	–	–	–	–
1.6.3 Handover to Ministry of Health	R	–	–	–	–	–	–	–	–	–	–	–

NETWORK

Network information:

Activity	Description	Preceding Activity	Time
A	System Analysis & Planning	None	20 days
B	Design user interface and user experience (UI/UX)	A	7 days
C	Design database tables and relationships	A	7 days
D	Prepare Prototype and design review	B, C	13 days
E	Frontend development	D	8 days
F	Backend development	D	10 days
G	Implement patient registration system and appointment booking feature	E, F	12 days
H	system Testing & Validation	G	22 days
I	Deployment & Documentation	H	15 days
J	Project Closure	I	11 days



ES	ID	EF
SL	Description	
LS	DUR	LF

Critical path: A>C>D>F>G>H>I>J

RESOURCE ALLOCATION

ID	RES	DUR	ES	LF	SL	0	10	20	30	40	50	60	70	80	90	100	110
A	4	20	0	20	0	4	4										
B	1	7	20	27	0			1									
C	1	7	20	27	0			1									
D	4	13	27	40	0				4								
E	2	8	40	50	1					2							
F	2	10	40	50	0					2							
G	4	12	50	62	0						4						
H	9	22	62	84	0							9					
I	5	15	84	99										5			
J	7	11	99	110												7	
Total						4	4	2	4	4	6 4	4 9	9	9 5	5 7	7	
Available						9	9	9	9	9	9	9	9	9	9	9	

TIME PHASED BUDGET

ID	DUR	BUD /100	ES	LF	SL	0	10	20	30	40	50	60	70	80	90	100	110
A	20	120	0	20	0	75	45										
B	7	40	20	27	0			40									
C	7	15	20	27	0			15									
D	13	70	27	40	0				70								
E	8	80	40	50	1					80							
F	10	100	40	50	0					100							
G	12	120	50	62	0						120						
H	22	185	62	84	0							185					
I	15	100	84	99										100			
J	11	170	99	110												170	
Total		1000				75	45	55	70	180	120	185		100		170	
cumulative						75	120	175	245	425	545	730		830		1000	

RISK MANAGEMENT:

Scope Creep (Post-Launch Feature Demands)

Risk Description: Stakeholders request new features beyond originally defined requirements.

Contingency Plan:

• Establish a formal change-management process requiring sponsor approval for any new feature.
• Evaluate new feature requests based on priority, impact, cost, and resource availability.
• Create a product roadmap to plan for future enhancements in controlled phases.
• Avoid adding features in emergency mode; instead, schedule them for future release

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Misunderstanding of Requirements (Gaps Visible After Release)

Risk Description: Delivered system does not fully match user needs or expectations.

Contingency Plan:

• Organize post-launch workshops with students, doctors, and reception staff to identify gaps.
• Conduct gap analysis and prioritize fixes or enhancements.
• Release incremental updates addressing misunderstanding areas.
• Improve documentation and user guides to clarify system workflows.
• Implement structured feedback loops to avoid repeating the issue.

Low User Adoption

Risk Description: Students or patients may not use the application as expected.

Contingency Plan:

• Provide in-app tutorials and onboarding flows to simplify first-time usage.
• Collect user feedback through surveys and implement requested improvements in iterative releases.
• Collaborate with university administration to mandate or strongly encourage usage for clinical training.
• Introduce incentives such as priority booking, digital badges for students, or faster service for patients.

Privacy and Security Risks

Risk Description: Unauthorized access, data breaches, or misuse of patients' medical data.

Contingency Plan:

• Immediately perform security audits and penetration testing if any breach is suspected.
• Notify the Ministry of Health and affected users as required by policy and regulations.
• Patch vulnerabilities and update frameworks to address identified threats.
• Implement stronger authentication (2FA), role-based access control, and encryption upgrades if gaps appear.
• Train staff on secure usage practices and data-handling protocols.
• Maintain regular database backups with secure isolation.

Technical Issues

Risk Description: Application may face performance issues, bugs, crashes, or downtime after launch.

Contingency Plan:

• Establish a dedicated technical support and maintenance team to respond to incidents promptly.
• Implement logging, monitoring, and alerting systems to detect issues early.
• Schedule periodic updates and patches for bug fixes.
• Maintain backup servers and load-balancing to reduce downtime.
• Apply hotfix releases for urgent critical bugs.

Dependency on University Infrastructure

Risk Description: The system depends on hospital reception computers, local networks, or university servers. Failures can disrupt operations.

Contingency Plan:

• Switch to a cloud-based backup hosting environment if on-premise servers fail.
• Provide offline-compatible features or caching for essential workflows.
• Develop a manual fallback process for registering patients if reception systems go down.
• Work with university IT teams to create SLAs for uptime, maintenance windows, and hardware support.
• Regularly audit university network performance and update infrastructure where needed.
